

MATHEMATICS 1

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 1

- 1 A shop sells two items, item P and item Q.

In January, the price of item P was 25% higher than the price of item Q.

In February, the price of item P decreased by 20%, while the price of item Q increased by 10%.

In March, the price of item P increased by 15%, while the price of item Q decreased by 5%.

In March, the price of item P was £6.30 more than the price of item Q.

What was the price of item Q in January?

- A £42
- B £50
- C £60
- D £66
- E £70
- F £75
- G £600

- 2 The expression $\frac{3\sqrt{3} + 3}{\sqrt{3} - 1}$ can be written in the form $a + b\sqrt{3}$, where a and b are integers.

What is the value of $a + b$?

- A 0
- B 3
- C 6
- D 9
- E 18

- 3 A fair four-sided die has faces labelled 1, 2, 3, 4.

A fair six-sided die has faces labelled 1, 1, 2, 3, 4, 6.

Both dice are rolled simultaneously. Let a be the score on the four-sided die and b be the score on the six-sided die.

What is the probability that the quadratic equation

$$x^2 + ax + b = 0$$

has real roots?

- A $\frac{7}{24}$
- B $\frac{3}{8}$
- C $\frac{5}{12}$
- D $\frac{11}{24}$
- E $\frac{7}{12}$

- 4 A square of side length x has area A .

The square is modified to form the pentagon shown in the diagram below:

- the base length b is 50% greater than x ,
- the height h is 20% less than x ,
- a right-angled triangular corner with base $\frac{1}{3}b$ and height $\frac{1}{2}h$ is removed from the top-right corner.

What is the area of the resulting pentagon in terms of A ?

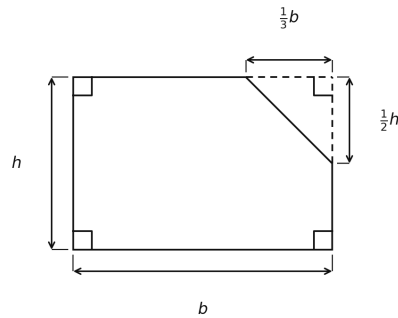


Diagram not to scale

- A $\frac{9}{10}A$
- B A
- C $\frac{11}{10}A$
- D $\frac{67}{60}A$
- E $\frac{6}{5}A$
- F $\frac{13}{10}A$

- 5 A component has a volume of V cubic inches. One inch is equivalent to L centimetres. The component is made from a material with a density of $D \text{ g/cm}^3$. The material is supplied in blocks, each with a mass of $M \text{ kg}$. Each block costs C pounds.

Which of the following expressions gives the cost of the material for the component, in pence? (There are 100 pence in one pound and 1000 grams in one kilogram.)

- A $\frac{VCDL^3}{10M}$
 B $\frac{VCDL}{10M}$
 C $\frac{VCD}{10ML^3}$
 D $\frac{100VCDL^3}{M}$
 E $\frac{VCDL^3}{1000M}$
 F $\frac{VCL^3}{10MD}$

- 6 The diagram shows a trapezium with parallel horizontal sides of length 2^{x+1} and 2^{x+3} , and perpendicular height 4^{2-x} .

Given that the area of the trapezium is $40\sqrt{2}$, what is the value of x ?

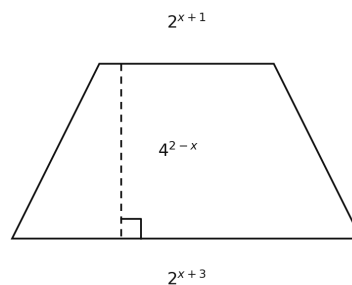


Diagram not to scale

- A -1.5
 B -0.5
 C 0.5
 D 1.5
 E 2.5

- 7 Let x and y be positive real numbers such that:

$$u = \log_2 \left(\frac{x^3 y^2}{32} \right)$$

and

$$v = \log_2 \left(\frac{xy^2}{8} \right)$$

Given that $u - v = 4$, what is the value of x ?

- A $\frac{1}{2}$
 - B 2
 - C 3
 - D 4
 - E 8
- 8 A 'twin-core' cable consists of two identical solid cylindrical metal wires running parallel to each other.
- The total mass of the metal in the cable is $7.2\pi \times 10^4$ kg and its density is 8.0×10^3 kg m⁻³.
- The cable is made of 2.5×10^7 segments joined end-to-end, each of length 2 cm.
- What is the radius of a single metal wire, in millimetres?
- A 0.3
 - B 3
 - C $3\sqrt{2}$
 - D 6
 - E $6\sqrt{2}$
 - F 9

- 9** A box contains only black discs and white discs. The total number of discs in the box is 15.
There are more black discs than white discs in the box.

Two discs are drawn at random from the box without replacement.

The probability that both discs drawn are of the same colour is $\frac{11}{21}$.

What is the probability that both discs drawn are black?

- A** $\frac{2}{21}$
- B** $\frac{1}{7}$
- C** $\frac{5}{21}$
- D** $\frac{2}{7}$
- E** $\frac{1}{3}$
- F** $\frac{3}{7}$
- G** $\frac{4}{9}$
- H** $\frac{2}{3}$

- 10** A set of 5 distinct integers has a mean of 10, a median of 5 and a range of 20.
The integer 8 is added to the set.

Which of the following correctly describes the changes to the mean, the median and the range of the set?

- A** Mean decreases, Median unchanged, Range increases
- B** Mean increases, Median increases, Range unchanged
- C** Mean decreases, Median increases, Range unchanged
- D** Mean increases, Median unchanged, Range increases
- E** Mean decreases, Median increases, Range increases

- 11** An unlimited supply of gold and silver coins is available. The probability of selecting a gold coin is $\frac{1}{3}$.

Two boxes, Box 1 and Box 2, are prepared. To prepare each box, two coins are selected from the supply and placed inside.

It is known that each box contains at least one gold coin.

One coin is then drawn from Box 1 and one coin is drawn from Box 2.

What is the probability that both coins drawn are gold?

- A** $\frac{1}{25}$
- B** $\frac{1}{9}$
- C** $\frac{9}{25}$
- D** $\frac{3}{5}$
- E** $\frac{9}{16}$
- F** $\frac{4}{5}$

- 12** A factory processes 1200 kg of raw material in batches of 20 kg.

In Stage 1, each batch is converted into 10 units of substance P and 5 units of substance G.

In Stage 2, all of substance P is processed, with each unit of P producing 0.5 units of substance G.

All the substance G from both stages is used to create a final product, F. 3 units of G are required to produce 2 units of F. The product F is produced at a constant rate of 5 units per minute.

What is the total time, in minutes, taken to produce all of F?

- A** 40
- B** 80
- C** 90
- D** 120
- E** 180
- F** 240

- 13** A composite cylindrical rod of length L consists of a solid inner core of radius R and a surrounding outer shell of inner radius R and outer radius $2R$.

The core is made of a material with density ρ and has mass M .

The shell is made of a material with density 2ρ .

What is the total mass of the rod?

- A** $3M$
- B** $4M$
- C** $6M$
- D** $7M$
- E** $8M$
- F** $9M$

- 14** A set of n distinct items ($n \geq 3$) is arranged in a random linear order. Two specific items, X and Y , are chosen from the set.

Which expression gives the probability that item X appears earlier in the arrangement than item Y ?

- A** $\frac{1}{n(n-1)}$
- B** $\frac{1}{n}$
- C** $\frac{2}{n}$
- D** $\frac{1}{2}$
- E** $\frac{n-1}{n}$

- 15** The quadratic equation in x

$$x^2 + (4^{2-2k} \times 8^k)x + 16^{2-k} = 0$$

has exactly one real root.

What is the value of k ?

- A** -3
- B** -2
- C** 0
- D** 1
- E** 2
- F** 3

- 16** Five years ago, the ratio of the age of P to the age of Q was 5 : 2. In 12 years, P will be 21 years older than Q.
What is the current age of P?

A 19
B 23
C 35
D 40
E 47

- 17** The variables y and x are related by the equation $y = \left(\frac{CL}{A}\right)x$, where C , L , and A are positive constants.

L is a length of 50 cm.

A is the area of a circular cross-section with a diameter of 2 mm.

The gradient of the graph of y against x is $\frac{1}{\pi} \times 10^{15}$.

What is the value of C in SI base units?

A 2×10^7
B 8×10^7
C 2×10^9
D 8×10^9
E 2×10^{11}
F 2×10^{13}

- 18** An exam paper is formatted in sections. For any integer $k \geq 1$, Section k contains exactly k pages.

The paper is printed such that every section must begin on an odd-numbered page. To achieve this, if a section ends on an odd-numbered page, a single blank page is inserted immediately after it. Section 1 begins on page 1.

What is the page number on which Section 51 begins?

A 1275
B 1276
C 1300
D 1301
E 1302
F 1326

- 19** A region in the xy -plane is defined by the following inequalities:

$$0 \leq x \leq a$$

$$0 \leq y \leq a$$

$$x^2 + y^2 \geq a^2$$

where a is a positive constant.

What is the area of this region?

- A** $a^2(1 - \frac{\pi}{4})$
- B** $\frac{\pi a^2}{4}$
- C** a^2
- D** $a^2(1 - \frac{\pi}{2})$
- E** $a^2(1 - \pi)$
- F** $\frac{\pi a^2}{2}$

- 20** An equilateral triangle and a regular 12-sided polygon are inscribed in the same circle. The area of the equilateral triangle is $27\sqrt{3}$.

What is the area of the regular 12-sided polygon?

- A** 27
- B** 54
- C** 108
- D** $108\sqrt{3}$
- E** 216
- F** 324

- 21** A cubic polynomial $P(x)$ with leading coefficient 1 can be factorised as $P(x) = (x - r_1)(x - r_2)(x - r_3)$, where r_1, r_2 and r_3 are positive integers.

The curve $y = P(x)$ passes through the point $(3, -4)$.

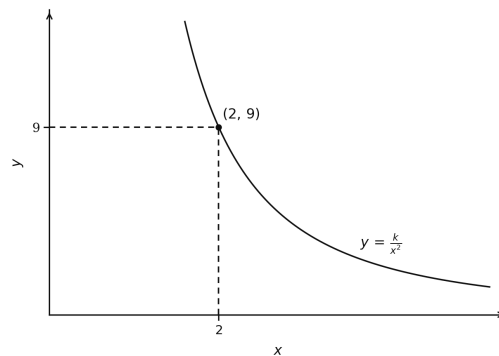
How many distinct polynomials $P(x)$ satisfy these conditions?

- A** 1
- B** 3
- C** 4
- D** 5
- E** 6
- F** 18

- 22** The diagram shows part of the graph of the curve $y = \frac{k}{x^2}$, where k is a positive constant and $x > 0$. The curve passes through the point $(2, 9)$.

The variable x is directly proportional to z^3 . When $z = 2$, $x = 24$.

Which of the following is an expression for y in terms of z ?



- A** $y = \frac{1}{4z^6}$
- B** $y = \frac{4}{z^6}$
- C** $y = \frac{4}{z^5}$
- D** $y = \frac{6}{z^3}$
- E** $y = \frac{12}{z^6}$
- F** $y = \frac{36}{z^6}$
- G** $y = \frac{81z^6}{4}$
- H** $y = 4z^6$

- 23** A system's success depends on two independent paths, A and B. The system succeeds if at least one path succeeds. Each path consists of 3 relays in series. A path succeeds only if all 3 of its relays succeed. All relays are identical and independent, each with a probability p of success. In 128 independent trials, the expected number of system successes is 30.

What is the value of p ?

- A** $\frac{1}{24}$
- B** $\frac{1}{8}$
- C** $\frac{3}{8}$
- D** $\frac{1}{2}$
- E** $\frac{5}{8}$
- F** $\frac{7}{8}$

- 24** Three regular polygons meet at a single vertex, completely covering the angle around it. One polygon is a square. The other two have n sides and m sides respectively, where $n < m$. What is the maximum possible value of m ?

- A** 8
- B** 12
- C** 16
- D** 20
- E** 24
- F** 42

- 25** Three quantities x , y and z are related by the ratios $x : y = 3 : 2$ and $z : y = 5 : 4$. What is the ratio $x : z$?

- A** 3 : 5
- B** 5 : 6
- C** 6 : 5
- D** 8 : 15
- E** 15 : 8

- 26** The variables x and y are positive.

For which of the following relationships is y inversely proportional to x ?

- A** The sum $x + y$ is a constant.
- B** $y = \frac{k}{x^2}$ for some non-zero constant k .
- C** $y = \frac{k}{x + c}$ for some non-zero constants k and c .
- D** The product xy is a non-zero constant.
- E** The ratio $\frac{y}{x}$ is a non-zero constant.
- F** $y = \frac{k}{x} + c$ for some non-zero constants k and c .

- 27** The following four lengths are denoted by P, Q, R, and S.

$$P = 5\mu\text{m}$$

$$Q = 0.02\text{mm}$$

$$R = 200\text{nm}$$

$$S = 5 \times 10^{-5}\text{m}$$

Arrange these lengths in order, from smallest to largest.

- A** R, P, Q, S
- B** S, Q, P, R
- C** P, Q, S, R
- D** Q, S, R, P
- E** R, P, S, Q
- F** R, Q, S, P